

MAT 322 HW01

Aditya Pahuja

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Problem 1. Prove the *Cauchy-Schwarz* inequality: for any two vectors $x, y \in \mathbb{R}^n$,

$$|\langle x, y \rangle| \leq \|x\| \|y\|.$$

Solution. Since both sides of the inequality are nonnegative, squaring gives the equivalent inequality

$$|\langle x, y \rangle|^2 \leq \|x\|^2 \|y\|^2.$$

Expanding in terms of the components tells us that we need to prove

$$(x_1 y_1 + x_2 y_2 + \cdots + x_n y_n)^2 \leq (x_1^2 + x_2^2 + \cdots + x_n^2)(y_1^2 + y_2^2 + \cdots + y_n^2)$$

for any $x_i, y_i \in \mathbb{R}$, $i = 1, 2, \dots, n$. The left-hand side is equal to

$$\sum_{i=1}^n (x_i y_i)^2 + 2 \sum_{1 \leq i < j \leq n} x_i y_i \cdot x_j y_j,$$

while the right-hand side is equal to

$$\sum_{i=1}^n x_i^2 y_i^2 + \sum_{1 \leq i < j \leq n} x_i^2 y_j^2 + x_j^2 y_i^2.$$

Noting that the first addends in the two expressions are the same, the inequality reduces to proving that

$$2 \sum_{1 \leq i < j \leq n} x_i y_i \cdot x_j y_j \leq \sum_{1 \leq i < j \leq n} x_i^2 y_j^2 + x_j^2 y_i^2.$$

Indeed, for any real numbers a, b, c, d , we have

$$a^2 b^2 + c^2 d^2 \geq 2abcd \iff a^2 b^2 - 2abcd + c^2 d^2 = (ab - cd)^2 \geq 0,$$

which is true because squares of real numbers are nonnegative (alternatively, this is just two-variable AM-GM). Thus, taking $(a, b, c, d) = (x_i, y_j, x_j, y_i)$ gives us the inequality

$$x_i^2 y_j^2 + x_j^2 y_i^2 \geq 2x_i y_j x_j y_i.$$

Summing over all pairs (i, j) with i, j integers satisfying $1 \leq i < j \leq n$ therefore yields the inequality

$$\sum_{1 \leq i < j \leq n} x_i^2 y_j^2 + x_j^2 y_i^2 \geq \sum_{1 \leq i < j \leq n} 2x_i y_j x_j y_i = 2 \sum_{1 \leq i < j \leq n} x_i y_i \cdot x_j y_j,$$

which is what we wanted. □

Problem 2. Prove the following statements:

- (a) For any $r > 0$, the r -ball around x , defined as

$$B_r(x) = \{y \in \mathbb{R}^n : \|x - y\| < r\},$$

is an open set.

- (b) A function $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is continuous at $x \in \mathbb{R}^n$ if and only if

$$\lim_{y \rightarrow x} f(y) = f(x).$$

- (c) If $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous at $x \in \mathbb{R}^n$ and $f(x) \neq 0$, then $1/f$ is a continuous function at x .
- (d) If $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is continuous at $x \in \mathbb{R}^n$, then its norm $\|f\|: \mathbb{R}^n \rightarrow \mathbb{R}$ is a continuous function at x . Is the converse true?

Solution. (a) We wish to show that for any $p \in B_r(x)$, there is an open ball centered at p that is a subset of $B_r(x)$. Consider the ball $B := B_{r-\|x-p\|}(p)$. If $y \in B$, then

$$\|x - y\| \leq \|x - p\| + \|p - y\| < \|x - p\| + (r - \|x - p\|) = r,$$

where the first inequality is the triangle inequality and the second follows from the definition of open ball, since y being in the ball B means that its distance to the ball's center p is less than the radius $r - \|x - p\|$. That is, the distance between x and y is smaller than r , so $y \in B$ implies $y \in B_r(x)$ as well, so $B_r(x)$ is open.

- (b) The statement

$$\lim_{y \rightarrow x} f(y) = f(x)$$

means that for every $\varepsilon > 0$, there exists a $\delta > 0$ such that $0 < \|x - y\| < \delta$ implies $\|f(x) - f(y)\| < \varepsilon$. This is the exact same thing as continuity at x ; the edge case $\|x - y\| = 0$ is irrelevant because obviously $\|f(x) - f(y)\|$ would also be $0 < \varepsilon$.

- (c) Since f is continuous at x , there exists a α such that

$$\|x - y\| < \alpha \implies \|f(x) - f(y)\| < \frac{\|f(x)\|}{2}.$$

This in particular means that $\|f(y)\| \geq \frac{\|f(x)\|}{2}$ when $\|x - y\| < \alpha$, since otherwise

$$\|f(x)\| \leq \|f(x) - f(y)\| + \|f(y)\| < \frac{\|f(x)\|}{2} + \frac{\|f(x)\|}{2} = \|f(x)\|.$$

Define $C = \frac{\|f(x)\|^2}{2}$ and let $\varepsilon > 0$. Since f is continuous at x , there is a $0 < \delta'$ such that $\|f(x) - f(y)\| < C\varepsilon$ whenever $\|x - y\| < \delta'$. Define $\delta = \min(\alpha, \delta')$. Then,

$$\|x - y\| < \delta \implies \|f(x) - f(y)\| < C\varepsilon \leq \|f(x)\| \varepsilon,$$

and the latter inequality rewrites to

$$\left| \frac{1}{f(y)} - \frac{1}{f(x)} \right| < \varepsilon$$

upon division by $|f(x)f(y)|$, which proves continuity.

(d) Given $\varepsilon > 0$, since f is continuous at x , let $\delta > 0$ be such that

$$\|x - y\| < \delta \implies \|f(x) - f(y)\| < \varepsilon.$$

We want to prove that there exists a $\delta' > 0$ such that

$$\|x - y\| < \delta' \implies |||f(x)\| - \|f(y)\||| = |||f(x)\| - \|f(y)\||| < \varepsilon.$$

The triangle inequality says that

$$\|f(x)\| \leq \|f(y)\| + \|f(x) - f(y)\|, \quad \|f(y)\| \leq \|f(x)\| + \|f(y) - f(x)\|,$$

so we get

$$-\|f(x) - f(y)\| \leq \|f(x)\| - \|f(y)\| \leq \|f(x) - f(y)\|$$

which is equivalent to

$$|||f(x)\| - \|f(y)\||| \leq \|f(x) - f(y)\|.$$

This means we can actually just take $\delta' := \delta$ to find that

$$\|x - y\| < \delta \implies |||f(x)\| - \|f(y)\||| \leq \|f(x) - f(y)\| < \varepsilon.$$

This proves that $\|f\|$ is continuous at x , as, for any $\varepsilon > 0$, we have found a $\delta' > 0$ such that $y \in B_{\delta'}(x)$ implies $f(y) \in B_\varepsilon(x)$.

The converse statement, however, is false. For example, take $f: \mathbb{R} \rightarrow \mathbb{R}$ to be the function $f(x) = 1$ for $x \geq 0$ and $f(x) = -1$ for $x < 0$. The norm is the constant function that is always equal to 1, which is obviously continuous, but f isn't continuous because the interval $(0, \infty)$ is an open set whose pre-image $f^{-1}((0, \infty)) = [0, \infty)$ is not open. $\square$

Problem 3. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} \sin(x) & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

At which points in $\mathbb{R}$ is f continuous?

Solution. The function is continuous at integer multiples of π , i.e. the points at which $\sin(x) = 0$. To check that f is continuous at $x = n\pi$ for any $n \in \mathbb{Z}$, let $\varepsilon > 0$. Since $\sin$ is continuous on all of $\mathbb{R}$, there exists a $\delta > 0$ such that

$$|n\pi - y| < \delta \implies |\sin(n\pi) - \sin(y)| = |\sin(y)| < \varepsilon.$$

Then, $|\sin(y)| = \max\{|\sin(y)|, 0\} \geq |f(y)|$ (for $f(y)$ is either equal to 0 or $\sin(y)$) and $f(n\pi) = 0$ on account of $n\pi$ being irrational, meaning

$$|n\pi - y| < \delta \implies |f(n\pi) - f(y)| = |f(y)| \leq |\sin(y)| < \varepsilon.$$

We have therefore shown that the definition of continuity at x holds when $x = n\pi$ for an integer n .

Now we have to show that f is discontinuous at all other points. If x is not an integer multiple of π , then $\sin(x) \neq 0$, so pick $\varepsilon = |\sin(x)|/2 > 0$ and let $\delta > 0$ be arbitrary. Additionally, let $\delta' > 0$ be chosen so that $|x - y| < \delta'$ implies that $|\sin(x) - \sin(y)| < \varepsilon/2$,

which we can do by continuity of $\sin$. Since $\mathbb{Q}$ and its complement are both dense in $\mathbb{R}$, we can pick a rational q and an irrational r that lie inside the interval $(x - m, x + m)$, where m is the smaller of δ, δ' . Then, either $f(x) = 0$, in which case $|x - r| < m$ implies

$$|f(x) - f(r)| = |\sin(r)| > |\sin(x)| - |\sin(x) - \sin(r)| > 2\varepsilon - \varepsilon/2 > \varepsilon,$$

so f can't be continuous at x , or $f(x) = \sin(x)$, in which case $|x - q| < m$ but

$$|f(x) - f(q)| = |f(x)| = |\sin(x)| = 2\varepsilon > \varepsilon.$$

In either case we have found a point in the interval $(x - \delta, x + \delta)$ whose image under f is not within $(f(x) - \varepsilon, f(x) + \varepsilon)$, so f cannot be continuous at any x outside $\pi\mathbb{Z}$. $\square$

Problem 4. Consider the function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $f(0, 0) = 0$ and

$$f(x, y) = \frac{xy}{x^2 + y^4} \text{ if } (x, y) \neq (0, 0).$$

At which points in $\mathbb{R}^2$ is f continuous?

Solution. f is continuous on $\boxed{\mathbb{R} \setminus \{(0, 0)\}}$.

First, we show that f is continuous on $\mathbb{R} \setminus \{(0, 0)\}$. This is pretty simple: the $\mathbb{R}^2 \rightarrow \mathbb{R}$ functions $a(x, y) = x$ and $b(x, y) = y$ are very obviously continuous on $\mathbb{R}$ (if $U \subseteq \mathbb{R}$ is open, then $a^{-1}(U) = U \times \mathbb{R}$ and $b^{-1}(U) = \mathbb{R} \times U$ are open), and then we can use the fact that sums, products, and quotients of continuous functions are continuous (with the caveat that the divisor in a quotient has to be nonzero) to conclude that

$$f(x, y) = \frac{xy}{x^2 + y^4}$$

is continuous whenever $x^2 + y^4 \neq 0$, i.e. all of $\mathbb{R} \setminus \{(0, 0)\}$.

To show that f is discontinuous at $(0, 0)$, consider the sequence $a_n = (\frac{1}{n}, \frac{1}{n})$ for $n \in \mathbb{N}$. This sequence has limit $(0, 0)$ in $\mathbb{R}^2$, but

$$\lim_{n \rightarrow \infty} f(a_n) = \lim_{n \rightarrow \infty} \frac{1/n^2}{1/n^2 + 1/n^4} = \lim_{n \rightarrow \infty} \frac{n^2}{n^2 + 1} = 1 \neq f(0, 0)$$

which cannot hold if f is continuous at $(0, 0)$. $\square$